

Project Report: Sales Prediction Using Machine Learning

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1. Project Overview

This project aims to predict product sales based on advertising expenditures across three different media channels: **TV**, **Radio**, and **Newspaper**. Using historical data from the advertising.csv file, we developed a machine learning model to help businesses optimize their marketing budget and forecast future revenue.

2. Data Analysis & Exploration

2.1 Dataset Summary

The dataset consists of 200 observations with the following key statistics:

- **Average TV Spend:** \$147.04
- **Average Radio Spend:** \$23.26
- **Average Newspaper Spend:** \$30.55
- **Average Sales:** 15.13 units

2.2 Correlation Analysis

To understand which platform drives the most sales, we calculated the correlation coefficients:

- **TV vs. Sales: 0.90** (Extremely Strong positive correlation)
- **Radio vs. Sales: 0.35** (Moderate positive correlation)
- **Newspaper vs. Sales: 0.16** (Weak positive correlation)

Visual Evidence:

The heatmap below illustrates that TV advertising is the dominant factor in determining sales volume.

3. Machine Learning Model

3.1 Algorithm

We implemented **Multiple Linear Regression**, a supervised learning algorithm that models the relationship between multiple independent variables and a single dependent variable.

3.2 Feature Importance (Coefficients)

The model assigns a weight to each platform, showing the predicted increase in sales for every \$1 spent:

- **TV:** 0.0545
- **Radio:** 0.1009
- **Newspaper:** 0.0043

(Note: While TV has a higher overall impact due to total spend, Radio actually shows a high efficiency per dollar spent according to this specific model iteration.)

4. Model Evaluation

The model was tested on a hold-out set (20% of the data) to ensure accuracy.

Metric	Value	Interpretation
R ² Score	0.9059	The model explains 90.6% of the variance in sales.
MAE	1.27	On average, predictions are off by only 1.27 units.
RMSE	1.70	Standard deviation of the prediction errors.

Actual vs. Predicted Plot:

The closer the points are to the red dashed line, the more accurate the model. Our results show a very tight fit.

5. Conclusions & Recommendations

1. **Invest Heavily in TV:** With a 0.90 correlation, TV is the most reliable driver of sales.
2. **Optimize Radio:** Radio spend shows significant efficiency and should be maintained as a secondary channel.
3. **Reduce Newspaper Spend:** Due to its weak correlation (0.16), the budget allocated here would likely generate a better ROI if moved to TV or Radio.
4. **Model Reliability:** With an R^2 score of over 90%, this model is a robust tool for future budget planning.